IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 26 November 2025, accepted 23 January 2026, date of publication 29 January 2026, date of current version 9 February 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3659151

RESEARCH ARTICLE

EDDNet30: A Spatial Attention and Multi-Scale Fusion Model for Enhanced Eye Disease Classification With Explainable AI

MD. ASRAFUL SHARKER NIROB [1], SHOWMICK GUHA PAUL [1], ARPA SAHA [1], MD ASSADUZZAMAN [1], MD ALAMGIR KABIR [1], (Member, IEEE), ARIF MAHMUD [1], (Member, IEEE), ATIQ UR REHMAN [2], AND AMINE BERMAK [2], (Fellow, IEEE)

[1]Health Informatics Research Laboratory (HIRL), Department of Computer Science and Engineering, Daffodil International University, Dhaka 1216, Bangladesh
[2]College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar

Corresponding authors: Atiq Ur Rehman (atirehman@hbku.edu.qa) and Md Alamgir Kabir (kabir.cse@diu.edu.bd)

This work was supported by the Open Access Funding provided by Qatar National Library.

ABSTRACT Eye diseases are a major global cause of blindness and vision impairment, highlighting the vital need for accurate and early diagnosis to prevent further deterioration. Despite advancements in medical imaging, the retinal diseases classification using fundus images remains challenging due to the complexity and subtlety of visual features. This study aims to develop and validate EDDNet30, a novel 30-layer deep learning model, to improve the classification of eye diseases from fundus images. The model incorporates advanced architectural features, such as spatial attention and multi-scale fusion modules, to enhance diagnostic accuracy and robustness. To validate the proposed model, a diverse dataset of 5531 images comprising 9 different disease categories is collected from various sources has been utilized. To enhance image quality, several pre-processing approaches were applied, including histogram equalization, color space conversion and contrast adjustment, ensuring high-resolution final images. Furthermore, image data augmentation was employed to increase the dataset's count, enhancing the model's ability to generalize during training. The spatial attention module helps the model to concentrate on the most relevant regions of the images, while multi-scale fusion modules capture and integrate details at different scales, significantly improving classification performance. To evaluate its effectiveness, the model was compared to various transfer learning models. The results show that EDDNet30 consistently outperforms transfer-learning models with 95.29% accuracy tested on 10% split test data, demonstrating superior accuracy and robustness in eye disease classification. Furthermore, model interpretability was enhanced by employing various explainable AI methods, such as Grad-CAM, Grad-CAM++ and LIME to identify and emphasize key features influencing the decision-making process. EDDNet30 represents a promising advancement in automated ocular disease detection, offering improved diagnostic reliability for clinical use.

INDEX TERMS Eye disease, spatial attention, disease classification, deep learning, fusion model, eye disease classification.

I. INTRODUCTION

Eye diseases represent a broad category of conditions that affect the eyes and can lead to impairment or loss of vision. This group of diseases includes mild to severe vision impairments like myopia, diabetic retinopathy, glaucoma, and age-related macular degeneration [1], [2], [3], [4], [5]. People of all ages can develop ocular diseases which are one of the most common causes of blindness worldwide and its risk increases with advancing age. The fact that about

The associate editor coordinating the review of this manuscript and approving it for publication was Giacomo Fiumara.

2.2 billion people have some degree of visual impairment or blindness puts into context just how prevalent ocular diseases actually are [6], [7], [8], [9]. Early detection of visual impairments can reduce blindness rates by 80% [10]. Better patient results and minimization of irreversible vision loss depend on timely recognition and correct diagnosis. Artificial intelligent systems, especially deep learning systems that have so far been developed and tested, are powerful tools toward improving the reliability as well as effectiveness in identifying eye diseases. State-of-the-art breakthroughs in deep learning have brought about a revolution in medical imaging, particularly by applying fundus images in the categorization of eye diseases. Deep learning, an emerging technology, has been widely applied in the modern era across various domains, effectively demonstrating its potential for accurate disease detection and other complex tasks [11], [12], [13], [14], [15], [16]. CNNs and transfer learning approaches are among the sophisticated models that have shown impressive accuracy in the diagnosis of eye diseases, particularly diabetic retinopathy, AMD, and glaucoma. These models use large datasets and high computational resources to learn complex patterns and features from fundus images sometimes surpassing human experts in some cases. However, despite their success, these models face several limitations. One major challenge is their ability to handle the subtle and complex features present in fundus images, such as variations in lighting, image quality, and disease manifestations. Moreover, numerous current models exhibit a deficiency in interpretability, hindering clinicians' trust and adoption of these systems in practical clinical environments [22], [23], [24]. The extensive use of automated diagnostic technologies in healthcare is significantly hampered by this lack of openness.

To address these limitations, this work introduces EDDNet30, a new 30-layer deep learning network, which enhances categorization of eye diseases based on fundus imaging. Unlike prior methods, EDDNet30 incorporates spatial attention mechanisms to focus on diagnostically relevant regions and multi-scale fusion modules to capture both fine- and coarse-grained features, to handle the subtlety and complexity of ocular pathology. These enhancements notably enhance the capability of the model to cope with high variability and complexity of fundus images, achieving a more accurate and robust classification. Also, in order to meet the interpretability, need, in this work we even leveraged the explainable AI methods (e.g., Grad-CAM, Grad-CAM++, and LIME) to the model. These strategies provide visual interpretations of the decisions underlying the model by highlighting the key features that most influence the expected outcomes. Such transparency is not only favorable for the trust in the model, however it also allows clinician explanations which leads to information which clinicians can act upon and results in improved diagnosis and treatment plan.

This study examines nine eye diseases alongside a normal class, analyzing their effects on vision and highlighting the vital importance of early diagnosis and treatment as illustrated in Figure 1.

A. CONTRIBUTIONS

This study introduces new deep learning techniques for automated eye disease classification, which hold significant important contributions for this field. The development of the EDDNet30 model, which integrates CNN with multi-fusion spatial attention mechanisms, enhances the accuracy and interpretability of fundus image analysis for a range of eye diseases. Key contributions include:

i. This study introduces EDDNet30, a novel deep learning model combining CNN with spatial attention mechanisms, optimized for fundus image analysis to improve feature encoding and classification accuracy.

ii. This study enhances the image quality using advanced pre-processing approaches, including histogram equalization, color space conversion and contrast adjustment, ensuring high-resolution images for improved model performance.

iii. This study implements advanced spatial attention mechanisms to focus on informative regions in retinal images, resulting in enhanced accuracy and interpretability of predictions.

iv. This study integrated explainable AI techniques, such as Grad-CAM++, and Grad-CAM and LIME to enhance EDDNet30's interpretability, identifying key features in its decision-making process for improved clinical reliability.

v. EDDNet30 achieves state-of-the-art performance, classifying 9 diverse eye diseases with 95.29% accuracy, paving the way for reliable real-world clinical deployment.

These advancements extend the current state-of-the-art in eye disease classification and highlight the effectiveness of attention mechanisms in managing complex medical imaging data.

The organization of this study is as follows: Section II reviews and details related works and recent advancements in eye disease classification. Section III details the methodology, including data collection, preprocessing techniques, and the architecture of the custom CNN multi-fusion spatial attention model. Section IV provides a comprehensive evaluation of experimental results and model performance. Finally, Section V summarizes the findings and discusses their broader implications for advancing automated disease diagnosis through medical imaging.

II. RELATED WORK

Eye diseases are a major concern in healthcare due to their potential to cause significant visual impairment and impact overall quality of life. The optimal treatment and management of these conditions necessitate their expeditious identification and precise diagnosis. In this context, machine learning (ML) and deep learning (DL) models have emerged as potential techniques to assist healthcare practitioners in the accurate diagnosis of ocular illnesses. This section will evaluate and analyze several studies on the subject, critically assessing and synthesizing existing research to

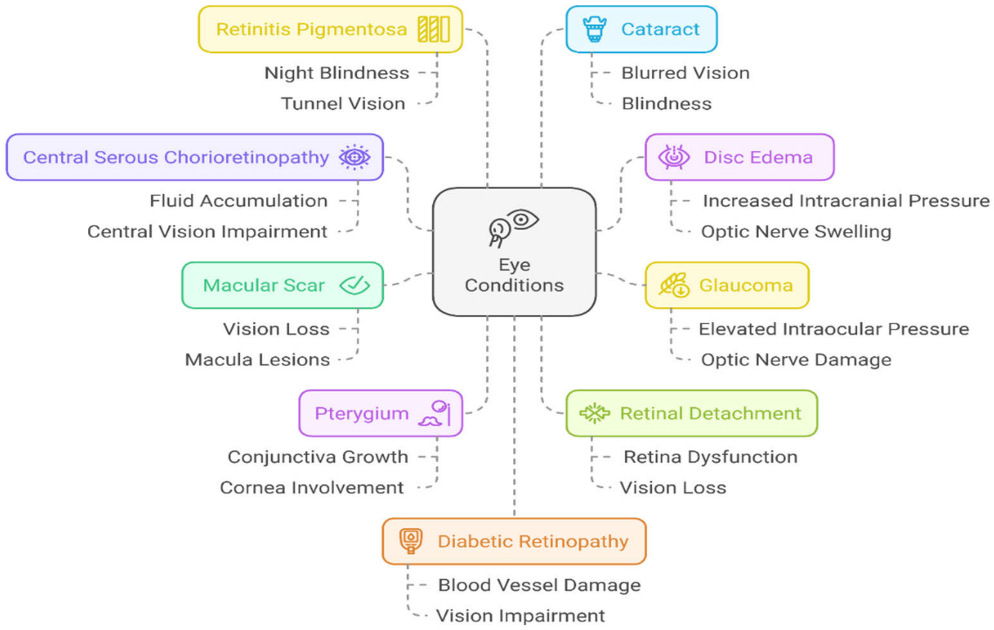

FIGURE 1. Map diagram of some common eye conditions with health issues.

give significant insights and context for future advances in automated eye disease identification. Recent studies have made impressive strides in using ML and DL for eye diseases detection from images like fundus images or optical coherence tomography (OCT) scans. Chowa et al. propose [25] OCCT, an optimized transformer-based model for the classification of eye diseases using an optical coherence tomography image. To address class imbalance, the study uses GAN-based data augmentation, increasing the dataset to 130,649 images. OCCT, refined through an ablation study, outperforms Vision Transformer (ViT), Swin Transformer, and CNN models including DenseNet, ResNet, MobileNetV2, achieving 97.09% accuracy. Extensive image preprocessing (morphological erosion, median filtering, alpha-beta correction) enhances quality. The model exhibits high accuracy despite limited training data, making it a robust and scalable AI tool for retinal disease detection.

Shoaib et al. [26] present the DiaGAN-CNN model, an advanced deep learning framework for diabetic retinopathy (DR) detection. This model combines GAN-based data augmentation with transfer learning, utilising InceptionResNet-v2 and Inception-v3 for feature extraction, to greatly improve classification accuracy. The study confirms its method with the Ocular Disease Intelligent Recognition (ODIR) dataset, reaching 98% accuracy and precision in identifying DR and other eye disorders. The DiaGAN-CNN model surpasses existing approaches by accurately collecting tiny retinal properties, resulting in high-speed and high-precision diagnoses.

Ma et al. [27] present the Intelligent Eye Multimodal Interactive Diagnostic System (IOMIDS), an AI-powered chabot that uses textual and image data for eye illness diagnosis and self-triage. In a two-stage cross-sectional research conducted across three medical centers, IOMIDS analyzed 50 disorders using 15,640 data entries. Among its multimodal models, the text + smartphone model had the greatest diagnosis accuracy (internal: 79.6%, external: 81.1%), beating the others. Multimodal integration enhanced answer quality and reduced disinformation, highlighting the use of AI Chabot's in automated ophthalmic care.

Malik et al. [28] introduce dual-branch semi-supervised learning for multi-class retinal disease classification using Color Fundus Photography (CFP). This approach integrates labeled and unlabeled data through self-training-based semi-supervised learning, improving diagnostic accuracy, especially in data-scarce environments. Evaluating six CNN models on the OIH dataset (4217 images), ResNet50 achieved the highest accuracy (93.14%), with 93% sensitivity, 98.37% specificity, and an AUC of 97.75%. The study provides insight of the potential of semi-supervised learning in clinical decision support systems (CDSS), making AI-powered ophthalmic diagnostics more accessible and effective.

Kansal et al. [29] present a DL-based system for automated ocular illness categorisation using the ODIR dataset (5,000 fundus images, 8 disease categories). The study uses transfer learning models (DenseNet201, EfficientNetB3, and InceptionResNetV2) and proposes a two-level feature selection method that combines Linear Discriminant Analysis (LDA) with neural network classifiers (DNN, LSTM, and BiLSTM). The BiLSTM classifier achieved 100% accuracy on training data and over 98% accuracy on validation data, demonstrating high efficiency and minimal processing complexity. The model provides a scalable, automated approach for clinical decision assistance, reducing ophthalmologists' workloads and enhancing early disease identification in resource-constrained environments.

Ali et al. [30] propose an automated framework aimed at assisting healthcare professionals in diagnosing ocular diseases, with AMD focus. The study enhances fundus images quality through preprocessing techniques such as contrast-limited adaptive histogram equalization and gamma correction to improve local contrast and feature intensity. The study introduces AMDNet23, a hybrid DL model that integrates CNN for feature extraction and LSTM mechanisms for feature analysis. This model achieves an impressive accuracy of 96.50% in AMD detection. The proposed model was benchmarked against 13 pre-trained CNN models, and it demonstrated superior performance. The dataset consisted of 2000 fundus images, which were categorized into four classifications. Nevertheless, the model's efficacy in larger and more diverse populations may be restricted by the relatively small dataset size, which is a significant limitation of this study. The model's applicability to a broader range of ocular conditions is restricted by its emphasis on a specific disease.

Fahdawi et al. [31] describe an automated deep learning technique, fundus-DeepNet system, which performs the multi-label categorization of different ocular diseases with the use of fundus images. The fundus images undergo several preprocessing steps which include contrast enhancement, circular border cropping, noise removal, and resizing. The data is also augmented. The system improves feature quality and robustness with the aid of deep learning blocks such as HRNet, attention, and SENet blocks. The use of a discriminative restricted Boltzmann machine with a softmax layer enables the detection of eight distinct ocular maladies. The dataset ophthalmic image analysis-ocular disease intelligent recognition was used to assess the model's performance. For the on-site and off-site test sets, the model achieved AUC, F1-scores, final scores, and Kappa scores which were all above 88% and 99% demonstrating its usefulness for ophthalmic diagnosis. Still, this model was tested primarily with the OIA-ODIR dataset, which is a limitation because it lacks the diversity and complexity of real-world clinical situations. The high number of deep learning blocks also contributed to increased model complexity which alongside model accuracy, is a limitation.

İbrahim and Çinar [32] conducted a study to investigate ML and DL approaches for automating the detection of ocular diseases, focusing specifically on eye diseases, diabetic retinopathy, and glaucoma. The study used and enlarged a dataset containing 4217 images, in which the pre-trained InceptionV3 model was used to recover 2048 features. With the 705 additional glaucoma and 370 diabetic retinopathy images, additional features were available for classification. The extracted features were classified with neural networks (NN), LR, KNN, and RF. The classifiers achieved the following accuracies before the dataset was augmented: 89.2% for NN, 87.3% for LR, 81.2% for KNN, and 76.9% for RF. After augmenting the dataset, the accuracies improved, with NN reaching 90.9%, LR 90.2%, KNN 84.6%, and RF 82%. The study reported recall, precision, F1 score, and AUC to evaluate the performance. One of the most relevant findings was the restriction in the dataset of 4217 images which was small and unbalanced and thus limited the generalizability of the trained model. There was also the concern regarding the accuracy improvements after augmentation. These were so substantial that they were deemed insufficient to capture the diverse presentation of ocular diseases, thus reducing the model's use in practice.

Aamir et al. [33] propose an adaptive threshold-based multi-level deep convolutional neural network detecting and classifying glaucoma from retinal fundus images. Glaucoma, a leading cause of blindness due to retinal damage, is traditionally diagnosed through lengthy, manual procedures. The study introduces a novel ML-DCNN architecture that consists of two phases: detection-net for identifying glaucoma, and classification-net for categorizing the severity of glaucoma into advanced, moderate, and early stages. The model's performance and competitiveness against state-of-the-art systems were demonstrated by its impressive results, which included an average sensitivity of 97.04%, accuracy of 99.39%, and precision of 98.2%, as a result of training on a dataset of 1,338 retinal images.

Shamsan et al. [34] propose a hybrid approach for automatic classification of the eye diseases from color fundus images using a combination of feature extraction and fusion techniques. Due to the difficulty in distinguishing between eye diseases at early stages, the study emphasizes the need for computer-assisted diagnostic methods. Three ways are examined: the initial strategy employs an ANN using features independently derived from DenseNet121and MobileNet models, with feature dimensionality decreased by principal component analysis. The second technique employs an artificial ANN on fused features from MobileNet and DenseNet121, both before to and after to feature reduction. The third way integrates ANN with the fused features from these models alongside handmade characteristics. The ANN model utilizing fused MobileNet and handmade features attained remarkable outcomes, with an AUC of 99.23%, accuracy of 98.5%, and sensitivity of 98.75%.

Wang et al. [35] tackle the issue of identifying ocular disorders from color fundus images by utilizing deep learning methodologies to enhance early detection and diagnostic precision. The study utilized a collection of 6,392 fundus pictures, then expanded to 17,766, to investigate five prominent CNN models EfficientNetB0, ShuffleNet, MobileNetV2, ResNet50, and ResNet101 alongside a bespoke 17-layer

CNN. Following the standardization of picture dimensions, the models were assessed using accuracy, Kappa coefficient, and precision metrics. Although ShuffleNet and EfficientNetB0 demonstrated commendable performance, the bespoke CNN surpassed all models, with 93% accuracy and a Kappa value of 92%. Moreover, the incorporation of picture characteristics with conventional machine learning classifiers, particularly logistic regression, enhanced the model's efficacy. The research illustrates the efficacy of AI in precisely identifying ocular illnesses and highlights the achievement of integrating bespoke convolutional neural networks with traditional techniques.

Sikder et al. [38] proposed an ensemble learning-based approach for the severity classification of diabetic retinopathy by extracting texture features and gray-level intensity from fundus images using the "APTOS 2019 BD" dataset. Their method demonstrated strong performance, achieving 94.20% accuracy and an F-measure of 93.51%, highlighting its potential as a reliable tool for large-scale retinal screening and early DR detection. This study reflects a significant advancement in traditional ML-based ophthalmic diagnostics through effective feature engineering and ensemble classification strategies.

Vadduri et al. [36] propose a DL-based approach for the segmentation and multi-class classification of diabetic eye disease (DED), a serious retinal condition common among diabetics. Early and precise detection is essential for preventing vision loss, and the study points out the roles of both quality and volume of fundus retinal images in building a reliable diagnostic system. The work used four pretrained models, ResNet50, Xception, VGG-16 and EfficientNetB7 and obtained better results with EfficientNetB7. Different image preprocessing operations, such as green channel extraction, CLAHE and illumination correction, were applied to enhance the image quality. Advanced segmentation techniques such as the Tyler coye algorithm, circular hough transform, and otsu thresholding were used to separate important anatomic features such as the optic nerve, blood vessels, and macula. The trained customized DCNN model on the preprocessed images achieved high accuracy for all tasks (cataract, diabetic retinopathy, glaucoma, and normal) with an accuracy of 96.43%, 98.33%, 97%, and 96%, respectively.

The study provides an extensive literature review of state-of-the-art in deep learning and machine learning techniques applied to eye diseases. Studies have taken into account various methods including pre-trained models, such as EfficientNet, ResNet, VCG, Inception, and hybrid CNN-LSTM models for feature extraction, which have been widely used for increasing classification accuracy. Table 1 highlights the related work summary. Moreover, methodologies including image enhancement, segmentation, and feature fusion have seen substantial performance improvements.

Despite these improvements, obstacles remain concerning dataset quantity, image quality, novel optimized models, and the necessity for explainability to ensure real-world application and generalization. Moreover, several studies concentrate exclusively on certain eye diseases, hence constraining the model's applicability in more extensive clinical environments where various conditions frequently occur. Numerous models encounter difficulties in addressing class imbalance, leading to biased predictions and reduced sensitivity. Although sophisticated architectures demonstrate promising outcomes, numerous models have difficulties in interpretability and transparency regarding their decision-making processes, hence impeding clinical implementation.

TABLE 1. Related studies summary.

Reference	Year	Sample size	Total class	Used models	Best model (accuracy)
Chowa et al. [25]	2025	130,649	4	CCT	97.09%
Shoaib et al. [26]	2024	4948	4	DiaGAN-CNN	98%
Ma et al. [27]	2025	15,640	8	Intelligent eye multimodal interactive diagnostic system	Internal: 79.6%, external: 81.1%
Malik et al. [28]	2025	4217	4	ResNet50	93.14%
Kansal et al. [29]	2025	5,000	8	BiLSTM	98%
Ali et al. [30]	2024	2000	4	AMDNet23	96.50%,
Fahdawi et al. [31]	2024	10,000	8	Fundus-DeepNet	Off-site: 92.41%, On-site: 92.66%
Kaya et al. [32]	2024	4217	4	Neural Network (NN)	90.9%
Aamir et al. [33]	2020	1338	3	multi-level DCNN	99.39%
Shamsan et al. [34]	2023	4217	4	MobileNet and handcrafted	98.5%
Wang et al. [35]	2023	5,000	8	MBSaNet	87.9%
Vadduri et al. [36]	2023	3948	4	DCNN	97%
REFUGE challenge winners [37]	2020	85,608	2	CNN	95.23%
Sikder et al. [38]	2021	2981	5	XGBoost	94.20%
Our study	2025	5531	10	EDDNet30	95.29%

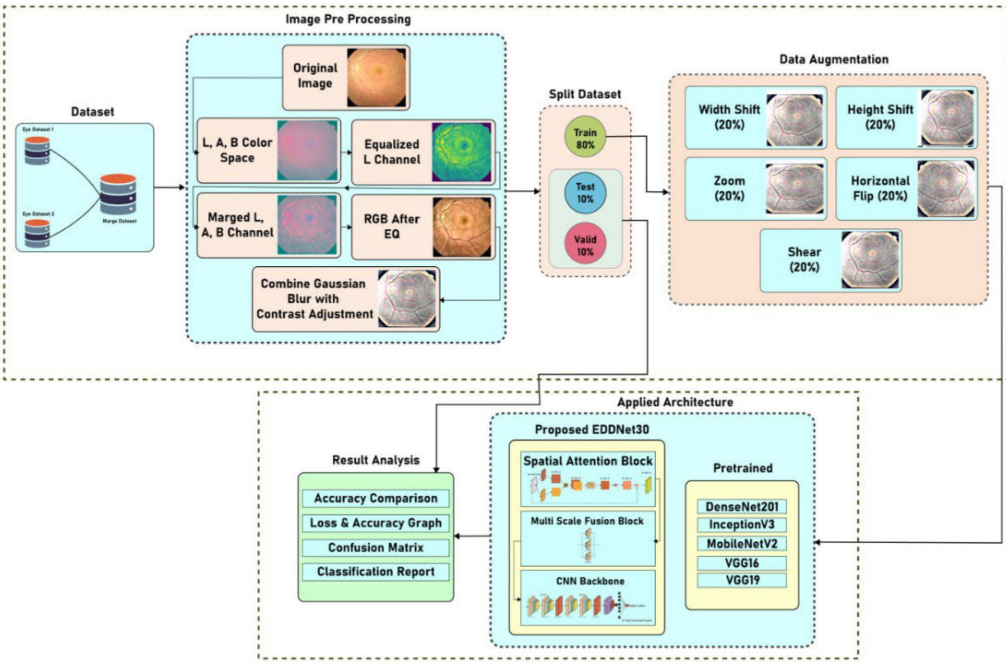

FIGURE 2. Comprehensive implementation procedure for eye disease detection.

This study enhances disease classification by utilizing several datasets that encompass 9 categories of eye diseases, hence boosting its significance and generalizability. This study presents a systematic image processing strategy designed to enhance feature highlighting, thereby facilitating the model's ability to identify patterns essential for accurate disease classification. This study incorporates a multi-scale and feature fusion-based CNN, presenting an innovative approach for the categorization of eye diseases. Eventually, this study presents a balanced and elucidated approach for the accurate, trustworthy, and clinically applicable diagnosis of eye diseases by explainable AI.

III. METHODOLOGY

Recent improvements in DL have transformed analysis of medical image, offering robust tools for automated evaluation and classification of visual data. In this study, an exhaustive methodology is implemented to accurately classify various eye diseases using deep learning approaches. At the core of this process is a structured framework, termed language data auditing, that encompasses several sequential steps beginning with the meticulous selection of diverse datasets from credible sources. These datasets span a wide array of pathological ocular conditions crucial for accurate modeling and diagnosis. Following this, several preprocessing techniques were used to increase the image quality and consistency. The datasets were aggregated, and data augmentation methods along with weight initialization were employed to enhance variety and address the class imbalance in the training data. A custom deep-learning model was then constructed and trained with the objective of achieving a comprehensive and high-performing 10-class classification. The methodology utilizes different architectural components, including pre-trained models for comparison and a custom deep learning model, EDDNet30 (eye disease detection network), introduced with novel spatial attention and multiscale fusion. This entire process is visually illustrated in Figure 2, providing a clear depiction of each step from data selection to model evaluation.

A. CONTRIBUTIONS DATA COLLECTION

The dataset for this study was constructed with a deliberate focus on capturing a wide range of clinically relevant eye diseases, allowing for comprehensive classification and diagnostic support. Data was carefully curated from various repositories, each contributing unique classes of eye. diseases that span a broad spectrum of pathological conditions. This diversity ensures that the resulting dataset represents a wide array of ocular diseases, thereby enhancing the model's ability to various different diagnostic categories generalization. Table 2 provides a comprehensive overview of the used dataset of this study, detailing the number of images for each of the 9 classes of eye diseases.

The dataset is composed of images from two primary repositories (eye dataset 1, eye dataset 2), each contributing unique classes of eye diseases [39], [40]. The figure 3 displayed the sample image of each eye disease class. To address the imbalanced among the datasets various augmentation techniques have been used, which ensures that

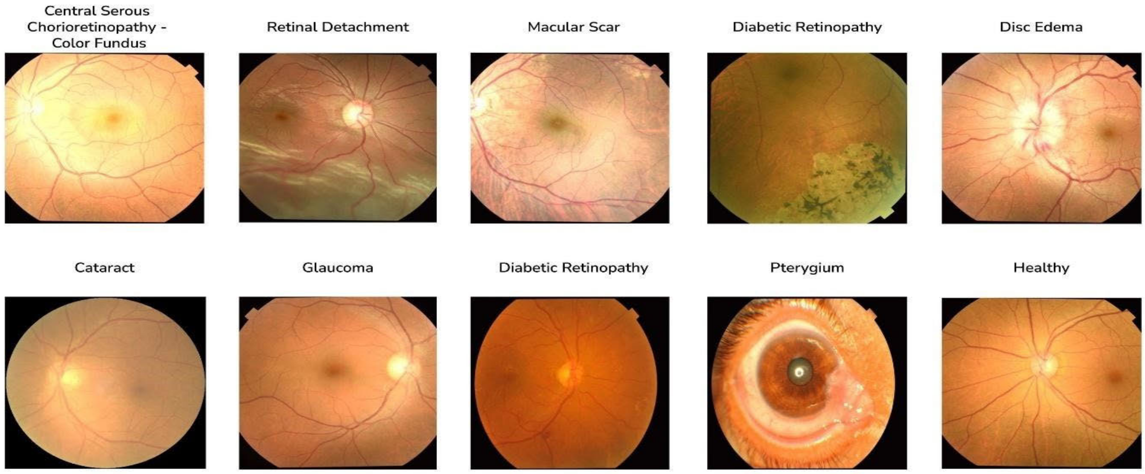

FIGURE 3. Visual representation of eye disease classes in the dataset.

TABLE 2. Dataset disease and image quantity distribution.

Dataset name	Class name	Image quantity
Eye dataset 1	Central serous chorioretinopathy - color fundus	101
	Diabetic retinopathy	1509
	Macular scar	444
	Disc edema	127
	Pterygium	17
	Retinitis pigmentosa	139
	Retinal detachment	125
	Healthy	1024
Eye dataset 2	Cataract	1038
	Glaucoma	1007
Total	**10 class**	**5531**

each disease class is adequately represented, contributing to more robust and generalized model performance.

- **Central serous chorioretinopathy (CSCR):** CSCR is defined by the accumulation of fluid beneath the retina, resulting in the detachment of retinal layers. This fluid build-up typically occurs in the central macular area, which is crucial for sharp, central vision. Patients with CSCR often experience distorted or blurred vision, a dark spot in the center of their visual field, and difficulty in perceiving colors. While many cases resolve spontaneously within a few months, chronic or recurrent cases may lead to permanent vision loss if untreated [41], [42], [43].
- **Disc edema:** Disc edema denotes the swelling of optic disc, the location where the optic nerve penetrates eye. Disc edema presents with symptoms such as blurred vision, headaches, and, in severe cases, vision loss. The presence of disc edema is a serious sign that warrants immediate investigation to determine the underlying cause, as it can indicate life-threatening conditions such as brain tumors or infections [44], [45].
- **Glaucoma:** Glaucoma includes a group of eye diseases characterized by damage to the optic nerve, in most cases associated with elevated intraocular pressure (IOP). Patients often do not notice symptoms until a considerable amount of damage has already taken place. Diagnosis is often based on routine eye examinations which include IOP measurements, optic nerve evaluation, and visual field testing [33], [46], [47], [48], [49].
- **Macular scar:** Macular scars are lesions that develop on the macula, the retinal region responsible for central vision. These scars may arise from several disorders, such as retinal detachment, age-related macular degeneration, or significant inflammation. Macular scars can lead to a permanent loss of sharp, detailed vision, making tasks like reading or recognizing faces difficult. The extent of vision impairment depends on the size and location of the scar [50], [51], [52].
- **Pterygium:** Pterygium is a benign growth of the conjunctiva that extends onto the cornea. This wing-shaped growth typically develops in response to prolonged exposure to ultraviolet (UV) light, wind, and dust, and is more common in individuals living near the equator. Pterygium can cause symptoms such as redness, irritation, and, in advanced cases, can obscure vision by encroaching on the visual axis. Although often asymptomatic, the condition can lead to astigmatism or block the pupil, causing significant visual impairment [53], [54], [55], [56].
- **Retinal detachment:** Retinal detachment is a serious disorder defined by the separation of light-sensitive layer of the retina at the posterior of the eye from its underlying supporting tissue. This separation impedes

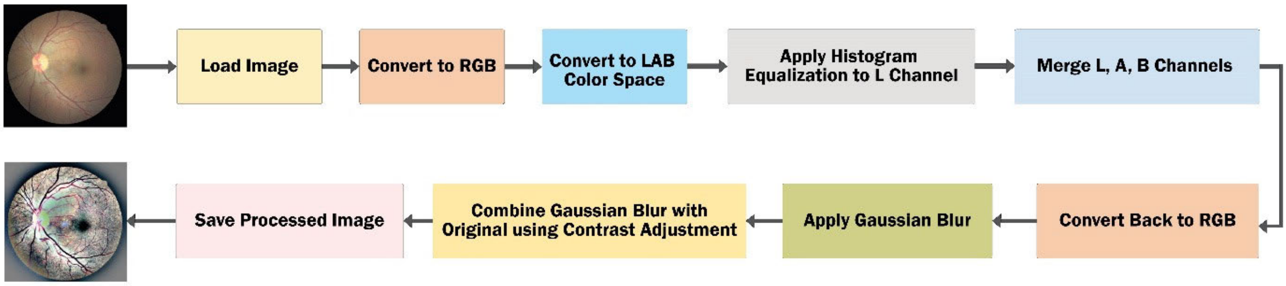

FIGURE 4. Comprehensive preprocessing workflow for eye disease detection.

the retina's functionality, resulting in visual impairment. Symptoms of retinal detachment encompass the abrupt emergence of floaters, a shadow or curtain-like impression obstructing the visual field and flashes of light. Untreated retinal detachment may lead to irreversible blindness in the afflicted eye.The most common causes include age-related changes, trauma, or diseases like diabetic retinopathy [57], [58], [59], [60].

- **Retinitis pigmentosa:** Retinitis pigmentosa encompasses a collection of rare, hereditary conditions characterized by the gradual degeneration of the retina. This degeneration results in night blindness, constricted peripheral vision, and ultimately, the loss of central vision. As the disease progresses, the photoreceptor cells (rods and cones) degenerate, leading to significant vision loss [61], [62].
- **Cataract:** Cataract is a prevalent age-related ocular condition resulting in visual impairment and marked by the opacification of the natural lens, and, if left untreated, potential blindness. The condition is primarily caused by aging but can also result from trauma, radiation exposure, or certain medications [63], [64], [65].
- **Diabetic retinopathy:** Diabetic retinopathy is a diabetes-related condition that affects the retinal blood vessels, causing visual impairment and increasing the chance of blindness. The illness is categories into two stages: no proliferative and proliferative [66], [67], [68]. Early symptoms may include blurred vision and floaters, but the condition often progresses without obvious signs.
- **Normal:** A normal eye functions as a sophisticated optical system that enables clear vision and effective visual processing. It comprises several key structures, including the lens, cornea, retina, and optic nerve, each contributing to its overall functionality.

B. DATA PRE-PROCESSING

Data preprocessing is an important step in DL-based eye disease detection, ensuring that the images are clean, standardized, and suitable for model training. Figure 4 outlines the preprocessing pipeline, consisting of various transformations aimed at improving the quality and consistency of input images.

The first step in this pipeline involves converting each image from its native format to the RGB color space to unify color representation across different sources. This is important because the dataset is sourced from multiple repositories, each having unique imaging parameters. Standardizing the images to a resolution of the pixels follows this RGB color conversion, ensuring uniform input size required by deep learning models. After resizing, the images are transformed into the Lab color space, where "L" represents luminance (or grayscale intensity), while "a" and "b" capture the color dimensions. A significant improvement implemented on the "L" channel is histogram equalization, a method that enhances visual contrast by reallocating pixel intensities. This modification accentuates essential visual characteristics, facilitating the model's ability to identify nuanced patterns linked to different ocular disorders. The mathematical expression for histogram equalization helps ensure optimal pixel intensity distribution across the image, leading to better feature visibility. The equation (1), (2) governing histogram equalization, *where* σ denotes the standard deviation of the gaussian function applied during convolution.

$$L_{eq}(i) = \frac{L(i) - L_{min}}{L_{max} - L_{min}} \times 255 \tag{1}$$

$$I_{blur}(x, y) = \frac{1}{2\pi\sigma^2} exp\,(-\frac{x^2 + y^2}{2\sigma^2}) \times I(x, y) \tag{2}$$

To further refine the images, gaussian blur is applied, as expressed by the gaussian convolution formula. This step reduces noise and smoothens the image, preparing it for contrast adjustment. The final contrast enhancement is achieved by combining the histogram-equalized image with the blurred image using a weighted combination. This results in sharper images with optimized contrast, making the critical features of the eye's anatomy more visible. The contrast adjustment equation is formulated as in equation (3) *where* $\alpha = 4, \beta = -4$ *and* $\gamma = 128$.

$$I_{final} = \alpha * I_{hist_{eq}} + \beta * I_{blur} + \gamma \tag{3}$$

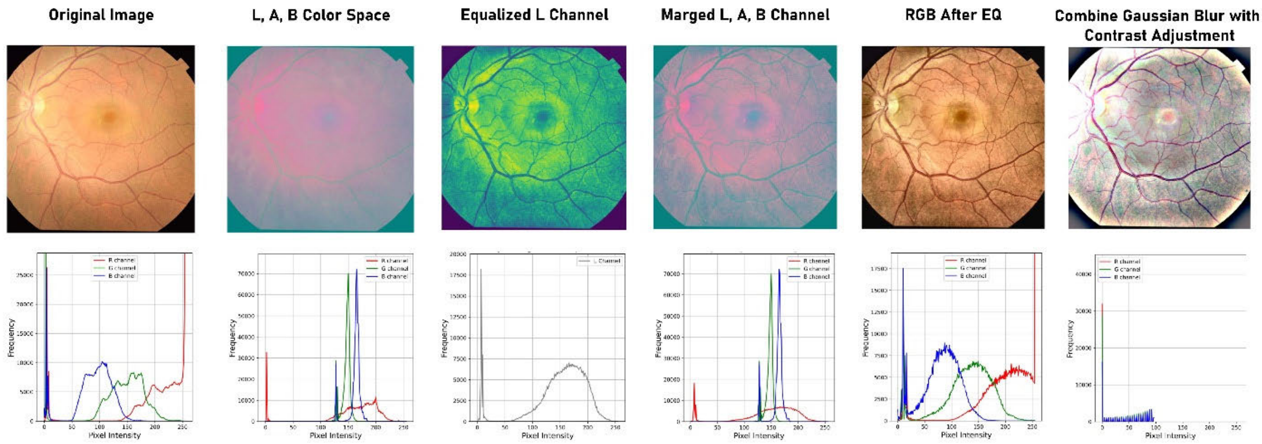

FIGURE 5. Retinal image processing for enhanced eye disease detection.

Once these transformations are completed, the processed images are stored in a dedicated directory for subsequent data augmentation and model training. This systematic approach ensures high-quality images, enabling the deep learning model to extract discriminative features effectively, which is essential for accurate detection of various eye diseases. Figure 5 illustrates a sequence of retinal image processing steps, crucial for eye disease detection. It begins with the original retinal image, showcasing visible blood vessels and the optic nerve. The next stage involves histogram equalization, enhancing contrast by adjusting pixel intensity distribution to bring out finer details such as vessels. Following this, a gaussian blur is applied, reducing noise and softening the image to highlight broader structures. Finally, the processed image results from multiple transformations, likely enhancing both large and small details. Below the processed images are color channel histograms (red, green, blue, and gray), which provide a detailed analysis of pixel intensity distributions in each channel, offering valuable insight into the image's color composition. This image-processing technique significantly enhances diagnostic precision in identifying retinal diseases, by accentuating vital structural features and minimizing extraneous details. It enriches the study by providing clearer visualization of pathological changes, facilitating more accurate and early detection of eye disorders.

C. TRANSFER LEARNING-BASED APPROCH

Transfer learning is a complex approach in which a model that has already been trained on a big dataset for one job is fine-tuned on a smaller dataset for a related task. This strategy improves performance greatly by using the model's capacity to transfer generalized characteristics learn in the original task, eliminating the requirement for extensive training on the smaller, domain-specific dataset. By fine-tuning these models, their unique attributes were harnessed to improve the system's accuracy and robustness. This method is particularly valuable in medical imaging, where annotated datasets are often limited. For eye disease detection, transfer learning allows the model to efficiently recognize critical features, such as blood vessel abnormalities and retinal lesions, which are crucial for identifying conditions. Consequently, transfer learning accelerates model development and enhances diagnostic precision, providing a more reliable and scalable solution for early detection of eye diseases.

- **DenseNet201:** DenseNet201 is a deep learning architecture that extends the notion of densely linked convolutional networks. The DenseNet model follows an efficient connection architecture, with each layer receiving input from previous levels and transmitting feature mappings to the next layer. This method addresses the vanishing gradient problem, enhances feature reuse, and decreases the number of parameters, resulting in more compact and efficient models. DenseNet201, with 201 layers, is a very deep variation designed to handle complicated picture classification and recognition tasks, making it ideal for applications like medical image analysis, object detection, and plant disease diagnosis. DenseNet201 improves learning performance by particularly in jobs that require feature separation. Its capacity to capture detailed characteristics while remaining computationally efficient makes it an effective tool for contemporary deep learning applications [69].
- **InceptionV3:** InceptionV3 is a highly advanced deep learning architecture that is part of the Inception series, designed for efficient and accurate image classification. It utilizes a unique module-based structure that enables the network to capture multi-scale features by performing convolutions of different sizes simultaneously within the same layer. This approach allows the model to extract rich, detailed features from images without significantly increasing computational complexity. One of the key innovations of InceptionV3 is the introduction of factorized convolutions, which break larger

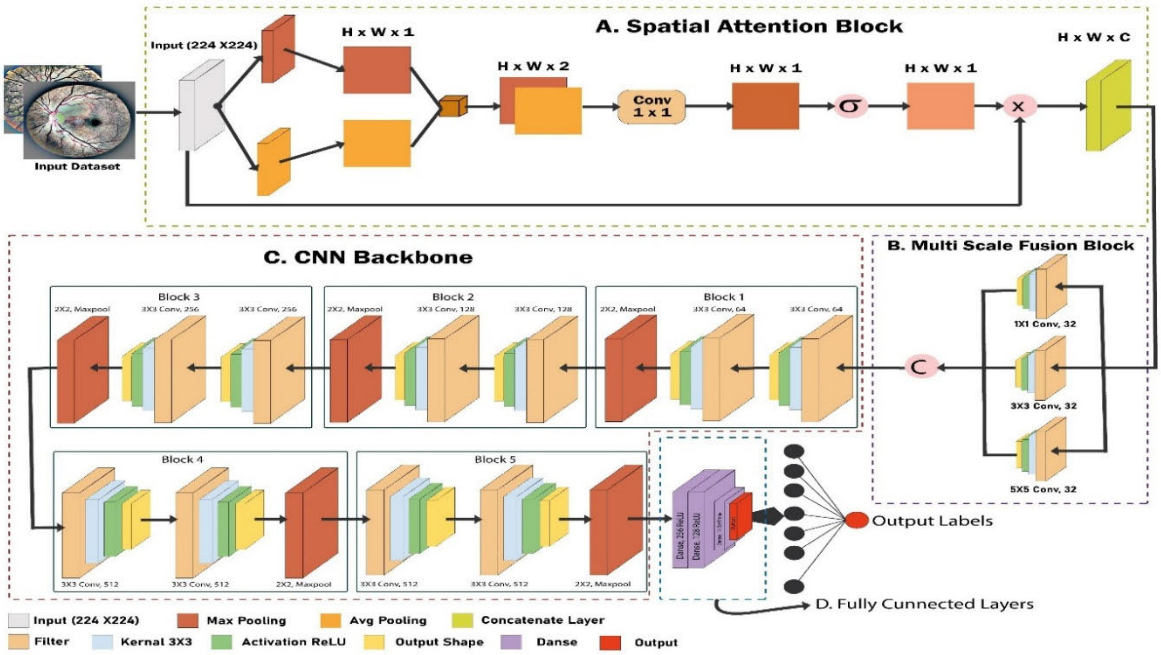

FIGURE 6. Architecture of the proposed EDDNet30.

convolutions into smaller, more efficient operations. This design decreases the amount of parameters while retaining excellent accuracy, resulting in a powerful yet lightweight InceptionV3. InceptionV3, which is widely utilized in computer vision applications, has been employed in a variety of domains, including medical imaging, object recognition, and plant disease detection. Its efficiency and accuracy make it an ideal candidate for large-scale and real-time image processing applications that require high performance and precision [70], [71].

- **MobileNetV2:** MobileNetV2 is deep learning lightweight model designed to perform efficiently on mobile and embedded devices without sacrificing accuracy. It builds upon the success of its predecessor, MobileNetV1, by introducing key architectural improvements. Depth-wise separable convolutions drastically reduce the number of parameters by breaking the standard convolution operation into two steps: depth-wise, which filters input channels independently and employs pointwise convolution to amalgamate them. The inverted residual structure ensures that the features are compressed (bottlenecked) and then expanded, allowing the network to efficiently capture spatial details with fewer computations. A linear layer at the bottleneck prevents the loss of important information. MobileNetV2 also features ReLU6 activation, which helps the model maintain a balance between precision and performance. These innovations make MobileNetV2 highly effective for tasks such as image classification, object detection, and segmentation, especially in resource-constrained environments [72], [73].
- **VGG16:** VGG16 is well-known deep CNN architecture for image categorization that is both simple and effective. VGG16 has 16 weight layers, including 13 convolutional layers and three fully linked layers. The architecture follows a straightforward design pattern where each convolutional layer uses small 3×3 filters, stacked in increasing depth as the network progresses. After the convolutional blocks, the network flattens the feature maps into a vector and passes it through the fully connected layers, culminating in a SoftMax activation for classification. VGG16's uniform use of 3×3 filters and its deep, yet manageable, layer depth make it both computationally efficient and highly effective for feature extraction, contributing to its success in various computer vision tasks [74].

D. PROPOSED APPROCH

This study introduces a novel methodology based on an enhanced deep learning model architecture designed to improve the accuracy and robustness of eye disease detection systems. The proposed model integrates spatial attention and multi-scale fusion blocks within a CNN framework, specifically optimized to capture and utilize critical features from eye disease images which has not been employed by previous studies [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38]. This advanced architecture facilitates more precise detection by focusing on the most relevant regions of the images, ensuring that subtle, yet significant, disease markers are not overlooked. The model achieves a depth of 30 layers, providing a highly sophisticated representation of the data.

The model architecture presented in fig. 6 outlines a deep CNN designed specifically for eye disease detection. This structure is optimized to capture critical features in retinal images, enabling precise identification of various eye diseases. The figure 6 outlines each layer's configuration and how they contribute to the overall functionality and performance of the model. The network begins with an input layer configured to accept images of shape (224, 224, 3), which is a standard input size for RGB images. Following this, a custom spatial attention layer is incorporated, which focuses on the most relevant areas of the input image by weighing the spatial importance of different regions. This is followed by the multi-scale fusion layer, another custom feature designed to combine information from multiple receptive field sizes, enabling the model to capture both fine and coarse details within the image.

The CNN is built with six convolutional blocks, each progressively increasing in complexity and depth. Convolution block 1 utilizes three convolutional layers with different kernel sizes (1×1, 5×5, and 3×3), each with 32 filters and ReLU activation. This multi-scale approach allows the model to capture features at various resolutions. The outputs of these convolutional layers are then concatenated to form a unified representation. Subsequent convolution blocks follow a more traditional CNN approach, with convolution blocks 2 to 6 consisting of two to three convolutional layers, all with ReLU activation and increasing filter sizes of 128, 256, 64, and 512, respectively. Each of these blocks is succeeded by a MaxPooling layer with a 2×2 pool size, which diminishes the spatial dimensions while most salient characteristics are preserved.

The model moves on to a flatten layer after the convolutional layers, which prepares it for the fully connected layers by converting the 2D feature mappings into a 1D vector. The ReLU function activates the two hidden layers with 256 and 128 units that make up the fully linked (dense) layers. The feature representations that the convolutional layers have learnt are further improved by these layers. Multi-class classification is finally made possible by the output layer's employment of a softmax activation function with 10 units, which correspond to 9 distinct classes or types of eye illnesses and one health eye classes. The design of EDDNet30 with 30 layers was determined through empirical ablation studies. We found this depth provided the optimal trade-off: shallower models (such as 18-layer) failed to capture the complexity of the 10-class problem, while deeper models (such as 50-layer) were prone to overfitting given our dataset size, without providing significant accuracy gains. The 30-layer architecture achieved the best balance between feature representation capacity and generalizability.

This sophisticated architecture leverages spatial attention, multi-scale feature extraction, and deep convolutional layers to enhance its capability in recognizing subtle and complex patterns in eye disease images, making it highly effective for diagnostic applications.

1) SPATIAL ATTENTION MECHANISM

The spatial attention mechanism enhances the performance of deep learning models by focusing on critical regions within input images [76]. This module includes a convolutional layer equipped with a sigmoid activation function to compute attention weights A, thereby dynamically adjusting the contribution of each pixel in the feature maps X. Mathematically, the spatial attention embedding is computed pointwise according to equation (4).

$$A = \sigma(W_a X + b_a) \tag{4}$$

where W_a represents the learnable weights and b_a denotes the biases of the convolutional layer. Here, σ denotes the sigmoid function applied element-wise across the spatial dimensions of X. The output A thus represents a map of weights that highlight regions of interest based on their relevance to the task at hand. The practical application of these attention weights involves performing an element-wise multiplication (Hadamard product) between A and X, resulting in equation (5).

$$Y = A \odot X \tag{5}$$

This operation effectively enhances the features associated with higher attention weights while diminishing those with lower weights. Consequently, the model becomes more proficient at identifying and emphasizing the discriminative features crucial for accurate eye disease detection.

2) MULTI-SCALE FUSION

The multi-scale fusion module enhances the model's ability to capture diverse and multi-level features by integrating convolutional layers with different kernel sizes: 1×1, 3×3, and 5×5. Each convolutional layer operates independently on the input feature maps X, extracting information at distinct spatial scales [77], [78], [79]. The resulting feature maps $F_1, F_{2,}$ and F_5 capture fine details, local patterns, and broader contexts respectively. To fuse these multi-scale features effectively, the module concatenates $F_1, F_{2,}$ and F_5 along the channel axis. This concatenation process forms a fused feature map F_{fusion}, which combines information from all three convolutional paths shown in equation (6).

$$F_{fusion} = Concatenate([F_1, F_2, F_5]) \tag{6}$$

This fused representation enhances the overall feature map F_{fusion} by incorporating comprehensive information across different spatial scales. By integrating multi-scale features, the model gains a deeper understanding of the input data, enabling it to identify intricate patterns and relationships essential for accurate eye disease detection. The module's design promotes versatility and adaptability, improving the model's robustness and performance in identifying subtle disease indicators across diverse cases in medical imaging and ophthalmology applications.

3) CNN BACKBONE

The proposed model's architecture comprises many convolutional layers integrated with max-pooling layers for spatial down sampling. The layers aim to extract hierarchical information from the enhanced picture input. Convolutional layers employ ReLU activation functions to introduce non-linearity into the feature extraction process.

Each convolutional layer i within the CNN backbone can be represented by equation (7).

$$F_i = ReLU(W_i \times F_{i-1} + b_i) \tag{7}$$

where F_{i-1} represents the input feature map, W_i denotes the learnable parameters (weights) specific to the $i-th$ convolutional layer, b_i represents the biases, $*$ denotes the convolution operation and ReLU denotes the rectified unit activation function.

4) FULLY CONNECTED LAYER

Following feature extraction, the flattened feature maps are processed through fully connected layers to perform detection. These layers facilitate learning of complex patterns in the feature representations extracted by the CNN backbone [80]. The fully connected layers are typically followed by SoftMax activation to output class probabilities. Mathematically, the computation of the output from the fully connected layers can be expressed as equation (8).

$$O = softmax(W_{fc} \times F_{fc} + b_{fc}) \tag{8}$$

where F_{fc} denotes the flattened feature vector from the CNN backbone, W_{fc} denotes the weights of the fully connected layer, b_{fc} denotes the bias, softmax denotes the softmax activation function applied element-wise to produce class probabilities.

IV. RESULT AND DISCUSSIOIN

This section contains the findings and discussions from a comprehensive review of various deep-learning models used to categories different eye disorders. The study focusses on a rigorous comparison of performance indicators such as precision, accuracy, sensitivity, specificity, and F1 score among models, including proven architectures and a unique hybrid model meant to improve performance. The research demonstrates how each model improves diagnosis accuracy for eye disease categorization.

A. PERFORMANCE COMPARISON

Table 3 provides a comparative analysis of model performances based on accuracy, recall, precision, and F1 score. VGG19, with 84.41% accuracy, outperforms VGG16 and MobileNetV2, which achieve lower accuracies of 84.13% and 82.63%, respectively. InceptionV3 shows moderate improvement, reaching 81.16% accuracy and a 0.81 F1 score. DenseNet201 marks further progress with 87.50% accuracy and a comparable F1 score of 0.87. However, EDDNet30 significantly outshines all others, achieving 95.29% accuracy, with near-perfect precision, recall, and F1 scores of 0.95, 0.95, and 0.95, solidifying its superior performance. Our comparative analysis (Table 3) reveals that EDDNet30 outperforms widely used transformer architectures like ViT and Swin Transformer on this task. This performance advantage is likely attributed to the inductive bias inherent in CNNs, which makes them more data-efficient than transformers when training on mid-sized medical datasets. Furthermore, the multi-scale fusion module in EDDNet30 allows for the simultaneous preservation of global context and fine-grained local features, such as minute hemorrhages or drusen, which can be challenging for patch-based transformer models to resolve without extensive pre-training on larger datasets.

TABLE 3. Performance metrics of various models.

Models	Accuracy	Precision	Recall	F1 Score
MobileNetV2	82.63%	0.83	0.83	0.83
VGG16	84.13%	0.86	0.85	0.85
VGG19	84.41%	0.85	0.84	0.84
InceptionV3	81.16%	0.81	0.81	0.81
DenseNet201	87.50%	0.87	0.87	0.87
Swin Transformer	94.10%	0.97	0.96	0.96
ViT	93.00%	0.94	0.93	0.93
Baseline(Pure CNN)	94.50%	0.95	0.95	0.95
Baseline+Spatial Attention	94.30%	0.96	0.96	0.96
Baseline+Multi-Scale Fusion	95.00%	0.95	0.95	0.95
EDDNet30	**95.29%**	**0.95**	**0.95**	**0.95**

B. EFFECT OF PREPROCESSING ON EDDNet30

Table 4 compares the performance of the EDDNet30 model with and without preprocessing across multiple eye disease datasets. Without preprocessing, the model achieves an average accuracy of 93.00% with precision, recall, and F1 scores of 0.90, 0.86, and 0.88, respectively. However, with preprocessing, the model demonstrates significant improvement, achieving a total average accuracy of 95.29%, with higher precision, recall, and F1 scores of 0.95, 0.95, and 0.95. Notably, preprocessing enhances performance in challenging categories, increasing accuracy and improves scores across key diseases, showing preprocessing's crucial role in boosting model effectiveness.

C. TRAINING PROGRESS OF EDDNet30

Figure 7(a) shows the EDDNet30 model's training and validation loss and accuracy graphs based on the original dataset. The model was trained using the Adamax optimizer with a learning rate of 0.001 and batch size of 32, using categorical crossentropy as the loss function and accuracy as the evaluation metric. To address the severe class imbalance within the dataset, the model was trained using focal loss

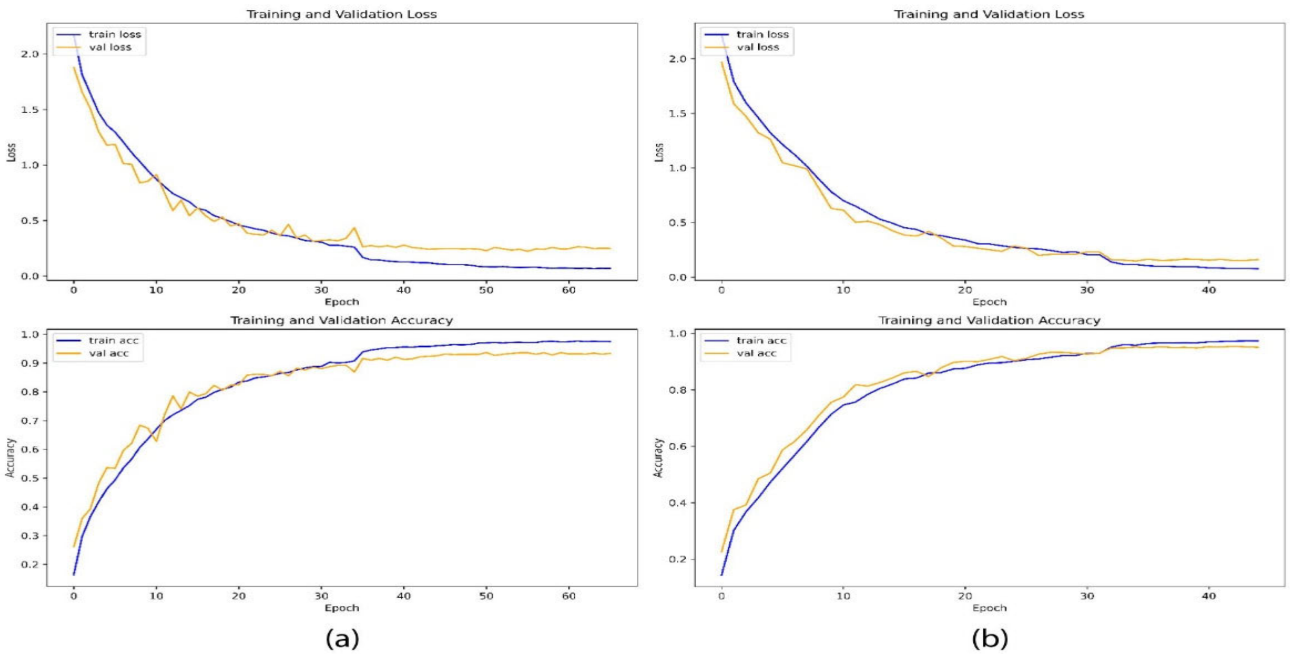

FIGURE 7. Training and validation performance of EDDNet30 on the original (a) and the pre-processed (b) dataset.

TABLE 4. Impact of preprocessing on EDDNet30 performance.

Models	Class name	Accuracy	Precession	Recall	F1 Score
EDDNet30 without pre-processing	Cataract	86.23%	0.90	0.86	0.86
	Central serous chorioretinopathy - Color Fundus	97.29%	0.95	0.97	0.96
	Diabetic retinopathy	88.24%	0.89	0.88	0.89
	Disc edema	93.96%	0.98	0.94	0.96
	Glaucoma	91.67%	0.86	0.92	0.89
	Macular scar	82.86%	0.85	0.83	0.84
	Healthy	90.14%	0.85	0.90	0.88
	Pterygium	100.00%	1.00	1.00	1.00
	Retinal detachment	98.03%	0.99	0.98	0.99
	Retinitis pigmentosa	97.22%	0.98	0.97	0.98
	Total Average	**93.00%**	**0.90**	**0.86**	**0.88**
EDDNet30 with pre-processing	Cataract	93.81%	0.89	0.93	0.91
	Central serous chorioretinopathy - Color Fundus	96.86%	0.96	0.97	0.96
	Diabetic retinopathy	87.64%	0.95	0.88	0.91
	Disc edema	99.35%	0.98	0.99	0.99
	Glaucoma	90.45%	0.94	0.90	0.92
	Macular scar	87.27%	0.89	0.87	0.88
	Healthy	95.89%	0.93	0.96	0.95
	Pterygium	100.00%	1.00	1.00	1.00
	Retinal detachment	99.27%	0.98	0.99	0.99
	Retinitis pigmentosa	99.21%	0.97	0.99	0.98
	Total Average	**95.29%**	**0.95**	**0.95**	**0.95**

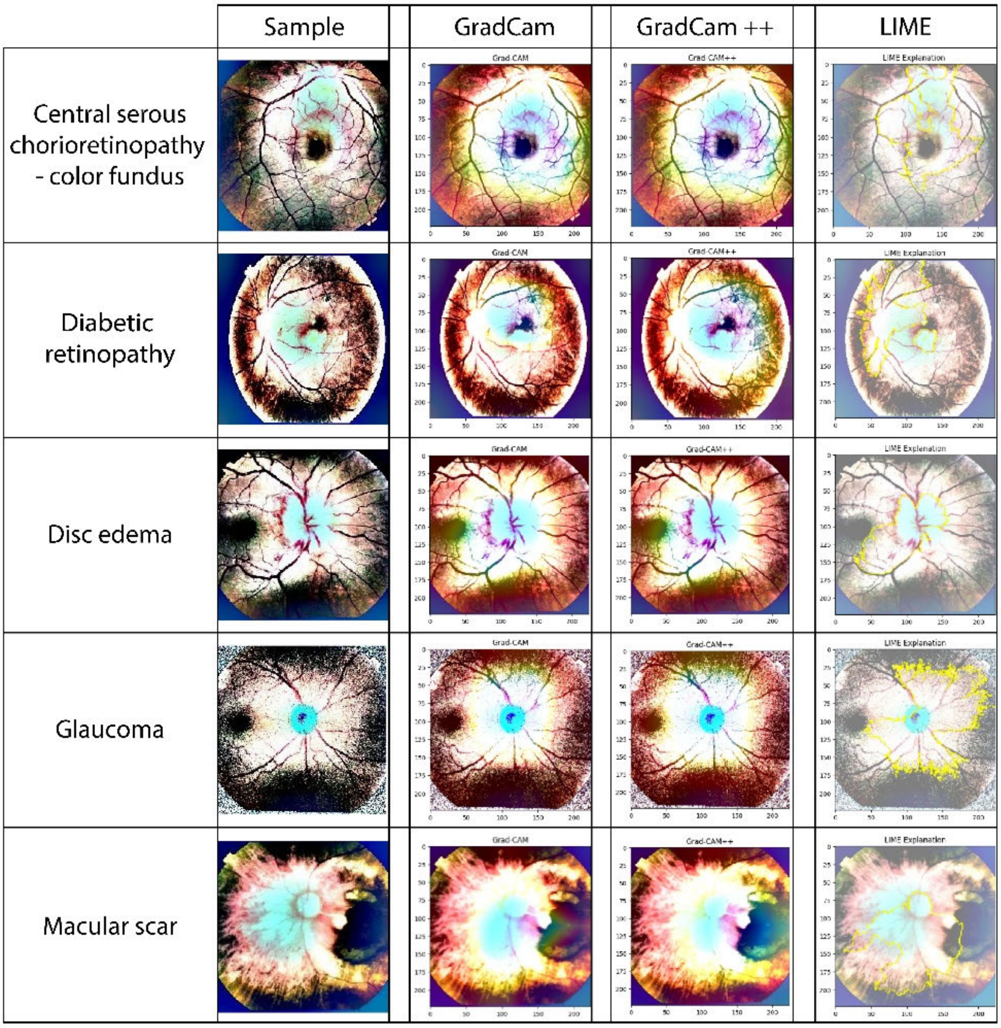

FIGURE 8. Example fundus images from 5 of the 10 classes with their corresponding XAI visualizations.

complemented by class weighting. Training was conducted on Kaggle using dual NVIDIA T4 GPUs (16GB each) with CUDA/cuDNN acceleration enabled. The training loss continuously reduces across 40 epochs, suggesting steady improvement in model learning, and the validation loss follows a similar decreasing pattern, albeit with small variations. On the accuracy side, the training accuracy steadily rises and plateaus near 0.95, signifying excellent model performance. The validation accuracy also increases, achieving a plateau at approximately 0.90, which is indicative of robust generalization. The model's ability to learn from the data without excessive variance is confirmed by the close alignment between training and validation metrics. Figure 7(b) illustrates the training and validation performance of the EDDNet30 model on the pre-processed dataset. The training loss consistently decreases over 40 epochs, demonstrating smooth learning progress, while the validation loss follows a similar downward trend with some minor fluctuations. The training accuracy shows a sharp increase, the proximity of the training and validation curves suggests no overfitting, indicating that the preprocessing phase substantially improves the model's generalization abilities and overall efficacy throughout the dataset, approaching near- perfect accuracy around 0.98, while the validation accuracy also rises steadily, stabilizing near 0.95. Figure 8 illustrates the visualization outcomes of grad-cam, grad-cam++, and lime applied to various eye disease specimens. The initial column presents the original fundus images, whereas the following columns explainability techniques. Grad-cam highlights the primary regions that have the greatest impact on a model's decision, providing basic essential feature localization. Grad-cam++ serves these visualizations with additional refined distributed precision, making it effective in quite complicated cases. This study employed LIME, or Local Interpretable Model Agnostic Explanations, in order to make the explanation of the prediction EDDNet30 model more interpretable, meaning that LIME provides local, instance-driven explanations.

LIME operates by modeling the vicinity of a given instance with a simpler, more interpretable model, allowing clinicians to comprehend the regions or features of the retinal images that motivated the model's prediction. Thus improves the clinician's trust in the model. The comparison among these techniques demonstrates their effectiveness in identifying pathological regions and enhancing model interpretability. The explanations and visualizations deepen understanding of depict the equivalent attention maps produced by the model's reasoning, thereby enabling trust regarding deep learning-based eye disease diagnosis.

D. COMPARATIVE ANALYSIS OF RELATED STUDIES

This section offers a detailed comparative analysis of state-of-the-art models in eye disease classification, highlighting performance metrics, dataset composition, and architectural trade-offs. As summarized in Table 5, various approaches demonstrate promising results, yet exhibit significant limitations when examined collectively. For instance, while Ali et al. [30] achieved 96.50% accuracy using AMDNet23 on 2,000 fundus images, the model focused solely on age-related macular degeneration and was limited by dataset size, constraining its generalizability across multiple disease types. Kaya et al. [32] obtained 90.9% accuracy through a neural network trained on a slightly expanded dataset, but issues such as class imbalance and susceptibility to overfitting persisted. Aamir et al. [33] reported an impressive 99.39% accuracy with ML-DCNN however, this was based on only 1,338 images and lacked diversity in disease classes, leading to questions about the model's robustness in real-world applications. Shamsan et al. [34] and Wang et al. [35] further pushed the boundaries of model accuracy, reaching 98.5% and 87.9%, respectively. Nonetheless, Shamsan's hybrid model involved high computational complexity, while Wang's approach suffered from lower generalizability despite employing a bespoke CNN. Vadduri et al. [36] introduced enhanced preprocessing and segmentation techniques that helped their DCNN model achieve 97% accuracy. However, the model's dependency on handcrafted features limited automation and scalability, thereby reducing its practicality in clinical environments.

In contrast to previous efforts, the new EDDNet30 model combines the benefits of existing solutions effectively overcoming their shared shortcomings. It attains 95.29% accuracy on a comprehensive dataset of 5,531 fundus images, which includes 9 distinct disease classes and one 'healthy' class. This model is much more representative and diverse than the datasets used in earlier studies.

The model also obtained a test accuracy of 96.36%, with a 95% confidence interval of 95.61% to 97.07%, demonstrating its consistent and reliable performance on unseen data. It is built to tackle several disease categories without being limited by sparse data and imbalanced classes. It achieves high accuracy and low computational cost, facilitating more practical healthcare deployment. Furthermore, EDDNet30 improves real-time applicability by removing the need for manual feature engineering through a fully automated CNN, streamlining the scaling process. With its robust design, targeted data augmentation, and optimised feature extraction, EDDNet30 ensures strong generalisation across complex multi-class classification tasks. This positions it as a more effective and adaptable solution in the domain of automated eye disease classification, with enhanced clinical relevance and operational efficiency.

TABLE 5. Similar studies comparison summary.

Reference	Year	Sample size	Total class	Used models	Best model (accuracy)
Chowa et al. [25]	2025	130,649	4	CCT	97.09%
Ali et al. [30]	2024	2000	4	AMDNet23	96.50%,
Kaya et al. [32]	2024	4217	4	Neural Network (NN)	90.9%
Aamir et al. [33]	2020	1338	3	multi-level DCNN	99.39%
Shamsan et al. [34]	2023	4217	4	MobileNet and handcrafted	98.5%
Wang et al. [35]	2023	5,000	8	MBSaNet	87.9%
Vadduri et al. [36]	2023	3948	4	DCNN	97%
REFUGE challenge winners [37]	2020	85,608	2	CNN	95.23%
Our study	2025	5531	10	EDDNet30	95.29%

V. CONCLUSION

The study demonstrates the significant change EDDNet30 an enhanced deep-learning model for accurate eye disease detection. Through integrating spatial attention and multi-scale fusion blocks into the CNN structure, EDDNet30 substantially strengthens the original network's capability to capture fine and complicated patterns and spatial relationships in eye images. Such elegant architecture achieved an impressive accuracy performance of 95.29%, which makes EDDNet30 one of the top models in eye disease classification. Evaluation using different performance measures always nominate EDDNet30 on top for a variety of eye disease classification tasks. It can also be observed from the result that the model has a high accuracy notably in distinguishing the type of various eye disease, an essential point for clinical applications. Moreover, the paper highlights the importance of data preprocessing for maximizing model performance, improve feature extraction, and noise reduction. These findings highlight the promising role of the novel architectural approach in the development of medical image analysis and automatic diagnostic systems. While our use of focal loss was effective in managing the dataset's significant imbalance, an exciting and important next step will be to test the model on a larger, more balanced dataset. This will allow us to further validate its strong performance and generalizability, especially for detecting rarer conditions. Future research should aim to refine these architectures, explore additional data augmentation techniques, and extend the model's applicability to diverse patient populations and clinical settings along with validating the model with external, multi-center datasets to overcome the limitation of internal testing and ensure real-world clinical generalizability. Furthermore, future investigations should prioritize a rigorous clinical evaluation involving ophthalmologists to assess the diagnostic relevance and reliability of the proposed XAI framework. The capabilities of EDDNet30 hold potential for improved accuracy in diagnosis and patient management and may provide guidance to future development of medical and deep learning techniques.

ACKNOWLEDGMENT

Open Access funding provided by the Qatar National Library.

Declaration of Generative AI Tools Use in the Writing Process:

Throughout the development of this manuscript, the authors utilized Generative AI tools for language enhancement. The authors reviewed and verified all content and take full responsibility for the final manuscript.

REFERENCES

[1] M. Fleckenstein, S. Schmitz-Valckenberg, and U. Chakravarthy, "Age-related macular degeneration: A review," *JAMA*, vol. 331, no. 2, p. 147, Jan. 2024, doi: 10.1001/jama.2023.26074.

[2] A. Cvekl and J. Vijg, "Aging of the eye: Lessons from cataracts and age-related macular degeneration," *Ageing Res. Rev.*, vol. 99, Aug. 2024, Art. no. 102407, doi: 10.1016/j.arr.2024.102407.

[3] Bhumika, N. S. Bora, and P. S. Bora, "Genetic insights into age-related macular degeneration," *Biomedicines*, vol. 12, no. 7, p. 1479, Jul. 2024, doi: 10.3390/biomedicines12071479.

[4] G. A. Wilke and R. S. Apte, "Complement regulation in the eye: Implications for age-related macular degeneration," *J. Clin. Invest.*, vol. 134, no. 9, May 2024, Art. no. 178296, doi: 10.1172/jci178296.

[5] S. Brodzka, J. Baszyński, K. Rektor, K. Hołderna-Bona, E. Stanek, N. Kurhaluk, H. Tkaczenko, G. Malukiewicz, A. Woźniak, and P. Kamiński, "The role of glutathione in age-related macular degeneration (AMD)," *Int. J. Mol. Sci.*, vol. 25, no. 8, p. 4158, Apr. 2024, doi: 10.3390/ijms25084158.

[6] *Vision Impairment and Blindness*. Accessed: Jul. 8, 2025. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

[7] *WHO Launches First World Report on Vision*. Accessed: Jul. 8, 2025. [Online]. Available: https://www.who.int/news/item/08-10-2019-who-launches-first-world-report-on-vision

[8] *One Billion People Have Preventable Eye Conditions, Increasingly Linked To Lifestyle Choices: UN Health Agency | UN News*. Accessed: Jul. 8, 2025. [Online]. Available: https://news.un.org/en/story/2019/10/1048762

[9] J. Drue. *At Least 2.2 Billion People Have Vision Impairment or Blindness, of Which Over 1 Billion Cases Could Have Been Prevented or Have Yet To Be Addressed*. Accessed: Jul. 8, 2025. [Online]. Available: https://lighthouseguild.org/at-least-2-2-billion-people-have-vision-impairment-or-blindness-of-which-over-1-billion-cases-could-have-been-prevented-or-have-yet-to-be-addressed/

[10] *Understanding the Leading Causes of Blindness*. Accessed: Jul. 8, 2025. [Online]. Available: https://ophthalmology.wustl.edu/understanding-the-leading-causes-of-blindness/

[11] M. Adil, A. Farouk, A. Ali, H. Song, and Z. Jin, "Securing tomorrow of next-generation technologies with biometrics, state-of-the-art techniques, open challenges, and future research directions," *Comput. Sci. Rev.*, vol. 57, Aug. 2025, Art. no. 100750, doi: 10.1016/j.cosrev.2025.100750.

[12] F. Naveed, A. Masih, J. Mahmood, M. Ahmed, A. Ali, A. Saddiqa, M. S. H. Abdulnabi, and E. Agbozo, "Sustainable AI for plant disease classification using ResNet18 in few-shot learning," *Array*, vol. 26, Jul. 2025, Art. no. 100395, doi: 10.1016/j.array.2025.100395.

[13] A.-A. Al-Maari, M. Abdulnabi, Y. Nathan, A. Ali, U. Ali, and M. Khan, "Optimized credit card fraud detection leveraging ensemble machine learning methods," *Eng., Technol. Appl. Sci. Res.*, vol. 15, no. 3, pp. 22287–22294, Jun. 2025, doi: 10.48084/etasr.10287.

[14] R. Ullah, N. Sarwar, M. N. Alatawi, A. A. Alsadhan, H. S. Alwageed, M. Khan, and A. Ali, "Advancing personalized diagnosis and treatment using deep learning architecture," *Frontiers Med.*, vol. 12, Mar. 2025, Art. no. 1545528, doi: 10.3389/fmed.2025.1545528.

[15] S. Hussain, N. Sarwar, A. Ali, H. Khan, I. Din, A. M. Alqahtani, M. Shabir, and A. Ali, "An enhanced random forest (ERF)-based machine learning framework for resampling, prediction, and classification of mobile applications using textual features," *Eng., Technol. Appl. Sci. Res.*, vol. 15, no. 1, pp. 19776–19781, Feb. 2025, doi: 10.48084/etasr.9148.

[16] M. Adil, A. Farouk, H. Abulkasim, A. Ali, H. Song, and Z. Jin, "NG-ICPS: Next generation industrial-CPS, security threats in the era of artificial intelligence, and open challenges with future research directions," *IEEE Internet Things J.*, vol. 12, no. 2, pp. 1343–1367, Jan. 2025, doi: 10.1109/JIOT.2024.3486659.

[17] World Health Organization. (Aug. 10, 2023). *Blindness and Vision Impairment*. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

[18] N. J. Shipley, "Setting our sights on vision: A rationale and research agenda for integrating eye-tracking into leisure research," *Leisure Sci.*, vol. 46, no. 3, pp. 274–295, Apr. 2024, doi: 10.1080/01490400.2021.1956389.

[19] F. Petrillo, A. Tortori, V. Vallino, M. Galdiero, A. M. Fea, U. De Sanctis, and M. Reibaldi, "Understanding acanthamoeba keratitis: An in-depth review of a sight-threatening eye infection," *Microorganisms*, vol. 12, no. 4, p. 758, Apr. 2024, doi: 10.3390/microorganisms12040758.

[20] J. A. Davidson, T. A. Ciulla, J. B. McGill, K. A. Kles, and P. W. Anderson, "How the diabetic eye loses vision," *Endocrine*, vol. 32, no. 1, pp. 107–116, Nov. 2007, doi: 10.1007/s12020-007-0040-9.

[21] K. Attebo, P. Mitchell, R. Cumming, and W. S. BMath, "Knowledge and beliefs about common eye diseases," *Austral. New Zealand J. Ophthalmology*, vol. 25, no. 3, pp. 283–287, Aug. 1997, doi: 10.1111/j.1442-9071.1997.tb01516.x.

[22] N. R. Galloway, W. M. K. Amoaku, P. H. Galloway, A. C. Browning, and N. R. Galloway, *Common Eye Diseases and Their Management*. Cham, Switzerland: Springer, 1999.

[23] M. M. Abdull, C. Chandler, and C. Gilbert, "Glaucoma, 'the silent thief of sight': Patients' perspectives and health seeking behaviour in Bauchi, northern Nigeria," *BMC Ophthalmology*, vol. 16, no. 1, p. 44, Dec. 2016, doi: 10.1186/s12886-016-0220-6.

[24] R. G. Mathew and I. E. Murdoch, "The silent enemy: A review of cataract in relation to glaucoma and trabeculectomy surgery," *Brit. J. Ophthalmology*, vol. 95, no. 10, pp. 1350–1354, Oct. 2011, doi: 10.1136/bjo.2010.194811.

[25] S. S. Chowa, M. R. I. Bhuiyan, I. J. Payel, A. Karim, I. U. Khan, S. Montaha, M. Z. Hasan, M. Jonkman, and S. Azam, "A low complexity efficient deep learning model for automated retinal disease diagnosis," *J. Healthcare Informat. Res.*, vol. 9, no. 1, pp. 1–40, Jan. 2025, doi: 10.1007/s41666-024-00182-5.

[26] M. R. Shoaib, H. M. Emara, A. S. Mubarak, O. A. Omer, F. E. Abd El-Samie, and H. Esmaiel, "Revolutionizing diabetic retinopathy diagnosis through advanced deep learning techniques: Harnessing the power of GAN model with transfer learning and the DiaGAN-CNN model," *Biomed. Signal Process. Control*, vol. 99, Jan. 2025, Art. no. 106790, doi: 10.1016/j.bspc.2024.106790.

[27] R. Ma et al., "Multimodal machine learning enables AI chatbot to diagnose ophthalmic diseases and provide high-quality medical responses," *NPJ Digit. Med.*, vol. 8, no. 1, pp. 1–18, Jan. 2025, doi: 10.1038/s41746-025-01461-0.

[28] M. H. Malik, Z. Wan, Y. Gao, and D.-W. Ding, "Efficient diagnosis of retinal disorders using dual-branch semi-supervised learning (DB-SSL): An enhanced multi-class classification approach," *Computerized Med. Imag. Graph.*, vol. 121, Apr. 2025, Art. no. 102494, doi: 10.1016/j.compmedimag.2025.102494.

[29] I. Kansal, V. Khullar, P. Sharma, S. Singh, J. A. Hamid, and A. J. Santhosh, "Multiple model visual feature embedding and selection method for an efficient ocular disease classification," *Sci. Rep.*, vol. 15, no. 1, pp. 1–29, Feb. 2025, doi: 10.1038/s41598-024-84922-y.

[30] M. A. Ali, M. S. Hossain, M. K. Hossain, S. S. Sikder, S. A. Khushbu, and M. Islam, "AMDNet23: Hybrid CNN-LSTM deep learning approach with enhanced preprocessing for age-related macular degeneration (AMD) detection," *Intell. Syst. Appl.*, vol. 21, Mar. 2024, Art. no. 200334, doi: 10.1016/j.iswa.2024.200334.

[31] S. Al-Fahdawi, A. S. Al-Waisy, D. Q. Zeebaree, R. Qahwaji, H. Natiq, M. A. Mohammed, J. Nedoma, R. Martinek, and M. Deveci, "Fundus-DeepNet: Multi-label deep learning classification system for enhanced detection of multiple ocular diseases through data fusion of fundus images," *Inf. Fusion*, vol. 102, Feb. 2024, Art. no. 102059, doi: 10.1016/j.inffus.2023.102059.

[32] K. İbrahim and İ. Çinar, "Evaluation of machine learning and deep learning approaches for automatic detection of eye diseases," *Intell. Methods Eng. Sci.*, vol. 1, no. 3, pp. 37–45, 2024, doi: 10.58190/imiens.2024.85.

[33] M. Aamir, M. Irfan, T. Ali, G. Ali, A. Shaf, A. S. S, A. Al-Beshri, T. Alasbali, and M. H. Mahnashi, "An adoptive threshold-based multi-level deep convolutional neural network for glaucoma eye disease detection and classification," *Diagnostics*, vol. 10, no. 8, p. 602, Aug. 2020, doi: 10.3390/diagnostics10080602.

[34] A. Shamsan, E. M. Senan, and H. S. A. Shatnawi, "Automatic classification of colour fundus images for prediction eye disease types based on hybrid features," *Diagnostics*, vol. 13, no. 10, p. 1706, May 2023, doi: 10.3390/diagnostics13101706.

[35] J.-Z. Wang, N.-H. Lu, W.-C. Du, K.-Y. Liu, S.-Y. Hsu, C.-Y. Wang, Y.-J. Chen, L.-C. Chang, W.-H. Twan, T.-B. Chen, and Y.-H. Huang, "Classification of color fundus photographs using fusion extracted features and customized CNN models," *Healthcare*, vol. 11, no. 15, p. 2228, Aug. 2023, doi: 10.3390/healthcare11152228.

[36] M. Vadduri and P. Kuppusamy, "Enhancing ocular healthcare: Deep learning-based multi-class diabetic eye disease segmentation and classification," *IEEE Access*, vol. 11, pp. 137881–137898, 2023, doi: 10.1109/ACCESS.2023.3339574.

[37] *REFUGE Competition Report*. Accessed: Jul. 8, 2025. [Online]. Available: https://public.grand-challenge-user-content.org/f/challenge/229/02053 2c2-cacf-48fc-97fa-ffb625a4d82d/REFUGE-VRT.pdf

[38] N. Sikder, M. Masud, A. K. Bairagi, A. S. M. Arif, A.-A. Nahid, and H. A. Alhumyani, "Severity classification of diabetic retinopathy using an ensemble learning algorithm through analyzing retinal images," *Symmetry*, vol. 13, no. 4, p. 670, Apr. 2021, doi: 10.3390/sym13040670.

[39] M. R. Rashid, S. Sharmin, T. Khatun, M. Z. Hasan, and M. S. Uddin, "Eye disease image dataset," Mendeley Data, V1, Apr. 2024, doi: 10.17632/s9bfhswzjb.1.

[40] G. V. Doddi. *Eye Diseases Classification*. Kaggle. Accessed: May 2025. [Online]. Available: https://www.kaggle.com/datasets/gunavenkatdoddi/ey e-diseases-classification

[41] M. Wang, I. C. Munch, P. W. Hasler, C. Prünte, and M. Larsen, "Central serous chorioretinopathy," *Acta Ophthalmologica*, vol. 86, no. 2, pp. 126–145, Mar. 2008, doi: 10.1111/j.1600-0420.2007.00889.x.

[42] M. Gemenetzi, G. De Salvo, and A. J. Lotery, "Central serous chorioretinopathy: An update on pathogenesis and treatment," *Eye*, vol. 24, no. 12, pp. 1743–1756, Dec. 2010, doi: 10.1038/eye.2010.130.

[43] R. Liegl and M. W. Ulbig, "Central serous chorioretinopathy," *Ophthalmologica*, vol. 232, no. 2, pp. 65–76, 2014, doi: 10.1159/000360014.

[44] S. Urfalioglu, G. Ozdemir, M. Guler, and G. G. Duman, "The evaluation of patients with optic disc edema: A retrospective study," *North. Clin. Istanb.*, vol. 8, no. 3, pp. 280–285, 2021. doi: 10.14744/nci.2020.25483.

[45] Y. Almog and M. Goldstein, "Visual outcome in eyes with asymptomatic optic disc edema," *J. Neuro-Ophthalmology*, vol. 23, no. 3, pp. 204–207, Sep. 2003.

[46] Q. Abbas, "Glaucoma-deep: Detection of glaucoma eye disease on retinal fundus images using deep learning," *Int. J. Adv. Comput. Sci. Appl.*, vol. 8, no. 6, pp. 1–15, 2017.

[47] D. A. Lee and E. J. Higginbotham, "Glaucoma and its treatment: A review," *Amer. J. Health-System Pharmacy*, vol. 62, no. 7, pp. 691–699, Apr. 2005, doi: 10.1093/ajhp/62.7.691.

[48] L. M. Nijm, J. Schweitzer, and J. Gould Blackmore, "Glaucoma and dry eye disease: Opportunity to assess and treat," *Clin. Ophthalmology*, vol. Volume 17, pp. 3063–3076, Oct. 2023, doi: 10.2147/opth.s420932.

[49] B. K. Betzler, S. M. Young, and G. Sundar, "Intraocular pressure and glaucoma in thyroid eye disease," *Ophthalmic Plastic Reconstructive Surgery*, vol. 38, no. 3, pp. 219–225, May 2022, doi: 10.1097/iop.0000000000002049.

[50] E. Daniel, C. A. Toth, J. E. Grunwald, G. J. Jaffe, D. F. Martin, S. L. Fine, J. Huang, G.-S. Ying, S. A. Hagstrom, K. Winter, and M. G. Maguire, "Risk of scar in the comparison of age-related macular degeneration treatments trials," *Ophthalmology*, vol. 121, no. 3, pp. 656–666, Mar. 2014, doi: 10.1016/j.ophtha.2013.10.019.

[51] P. G. Christakis, E. Agrón, M. L. Klein, T. E. Clemons, J. P. Campbell, F. L. Ferris, E. Y. Chew, and T. D. Keenan, "Incidence of macular atrophy after untreated neovascular age-related macular degeneration," *Ophthalmology*, vol. 127, no. 6, pp. 784–792, Jun. 2020, doi: 10.1016/j.ophtha.2019.11.016.

[52] D.-U. Bartsch, M. Intaglietta, J. F. Bille, A. W. Dreher, M. Gharib, and W. R. Freeman, "Confocal laser tomographic analysis of the retina in eyes with macular hole formation and other focal macular diseases," *Amer. J. Ophthalmology*, vol. 108, no. 3, pp. 277–287, Sep. 1989, doi: 10.1016/0002-9394(89)90118-9.

[53] J. C. Hill and R. Maske, "Pathogenesis of pterygium," *Eye*, vol. 3, no. 2, pp. 218–226, Mar. 1989, doi: 10.1038/eye.1989.31.

[54] S.-M. Saw and D. Tan, "Pterygium: Prevalence, demography and risk factors," *Ophthalmic Epidemiology*, vol. 6, no. 3, pp. 219–228, Jan. 1999, doi: 10.1076/opep.6.3.219.1504.

[55] J. Tan, U. Vollmer-Conna, L. Tat, and M. Coroneo, "Dry-eye disease in recurrent pterygium," *Ophthalmic Res.*, vol. 61, no. 4, pp. 199–203, 2019, doi: 10.1159/000493544.

[56] T. Shahraki, A. Arabi, and S. Feizi, "Pterygium: An update on pathophysiology, clinical features, and management," *Therapeutic Adv. Ophthalmology*, vol. 13, Jan. 2021, Art. no. 251584142110201, doi: 10.1177/25158414211020152.

[57] The Eye Disease Case-Control Study Group, "Risk factors for idiopathic rhegmatogenous retinal detachment," *Amer. J. Epidemiology*, vol. 137, no. 7, pp. 749–757, Apr. 1993, doi: 10.1093/oxfordjournals.aje.a116735.

[58] D. N. Shah, A. Al-Moujahed, C. W. Newcomb, R. O. Kaçmaz, E. Daniel, J. E. Thorne, C. S. Foster, D. A. Jabs, G. A. Levy-Clarke, R. B. Nussenblatt, J. T. Rosenbaum, H. N. Sen, E. B. Suhler, N. P. Bhatt, and J. H. Kempen, "Exudative retinal detachment in ocular inflammatory diseases: Risk and predictive factors," *Amer. J. Ophthalmology*, vol. 218, pp. 279–287, Oct. 2020, doi: 10.1016/j.ajo.2020.06.019.

[59] R. Amer, H. Nalcı, and N. Yalçındağ, "Exudative retinal detachment," *Surv. Ophthalmology*, vol. 62, no. 6, pp. 723–769, Nov. 2017, doi: 10.1016/j.survophthal.2017.05.001.

[60] H. Lewis, "Peripheral retinal degenerations and the risk of retinal detachment," *Amer. J. Ophthalmology*, vol. 136, no. 1, pp. 155–160, Jul. 2003, doi: 10.1016/s0002-9394(03)00144-2.

[61] F. Parmeggiani, F. S. Sorrentino, D. Ponzin, V. Barbaro, S. Ferrari, and E. Di Iorio, "Retinitis pigmentosa: Genes and disease mechanisms," *Current Genomics*, vol. 12, no. 4, pp. 238–249, Jun. 2011, doi: 10.2174/138920211795860107.

[62] C. Hamel, "Retinitis pigmentosa," *Orphanet J. Rare Diseases*, vol. 1, no. 1, p. 40, Dec. 2006, doi: 10.1186/1750-1172-1-40.

[63] K. Naderi, J. Gormley, and D. O'Brart, "Cataract surgery and dry eye disease: A review," *Eur. J. Ophthalmology*, vol. 30, no. 5, pp. 840–855, Sep. 2020, doi: 10.1177/1120672120929958.

[64] J. R. Chang, E. Koo, E. Agrón, J. Hallak, T. Clemons, D. Azar, R. D. Sperduto, F. L. Ferris, and E. Y. Chew, "Risk factors associated with incident cataracts and cataract surgery in the age-related eye disease study (AREDS)," *Ophthalmology*, vol. 118, no. 11, pp. 2113–2119, Nov. 2011, doi: 10.1016/j.ophtha.2011.03.032.

[65] M. Miura, T. Inomata, M. Nakamura, J. Sung, K. Nagino, A. Midorikawa-Inomata, J. Zhu, K. Fujimoto, Y. Okumura, K. Fujio, K. Hirosawa, Y. Akasaki, M. Kuwahara, A. Eguchi, H. Shokirova, and A. Murakami, "Prevalence and characteristics of dry eye disease after cataract surgery: A systematic review and meta-analysis," *Ophthalmology Therapy*, vol. 11, no. 4, pp. 1309–1332, Aug. 2022, doi: 10.1007/s40123-022-00513-y.

[66] E. J. Sussman, "Diagnosis of diabetic eye disease," *JAMA, J. Amer. Med. Assoc.*, vol. 247, no. 23, p. 3231, Jun. 1982, doi: 10.1001/jama.1982.03320480047025.

[67] A. J. Barber, "A new view of diabetic retinopathy: A neurodegenerative disease of the eye," *Prog. Neuro-Psychopharmacology Biol. Psychiatry*, vol. 27, no. 2, pp. 283–290, Apr. 2003, doi: 10.1016/s0278-5846(03)00023-x.

[68] S. Krishnaiah, T. Das, P. K. Nirmalan, B. R. Shamanna, R. Nutheti, G. N. Rao, and R. Thomas, "Risk factors for diabetic retinopathy: Findings from the Andhra Pradesh eye disease study," *Clin. Ophthalmology*, vol. 1, no. 4, pp. 475–82, 2007.

[69] S. Ovreiu, E.-A. Paraschiv, and E. Ovreiu, "Deep learning & digital fundus images: Glaucoma detection using DenseNet," in *Proc. 13th Int. Conf. Electron., Comput. Artif. Intell. (ECAI)*, Jul. 2021, pp. 1–4, doi: 10.1109/ECAI52376.2021.9515188.

[70] Y. Pan, J. Liu, Y. Cai, X. Yang, Z. Zhang, H. Long, K. Zhao, X. Yu, C. Zeng, J. Duan, P. Xiao, J. Li, F. Cai, X. Yang, and Z. Tan, "Fundus image classification using inception V3 and ResNet-50 for the early diagnostics of fundus diseases," *Frontiers Physiol.*, vol. 14, Feb. 2023, Art. no. 1126780, doi: 10.3389/fphys.2023.1126780.

[71] S. Faizal, C. A. Rajput, R. Tripathi, B. Verma, M. R. Prusty, and S. S. Korade, "Automated cataract disease detection on anterior segment eye images using adaptive thresholding and fine tuned inception-v3 model," *Biomed. Signal Process. Control*, vol. 82, Apr. 2023, Art. no. 104550, doi: 10.1016/j.bspc.2022.104550.

[72] R. Patel and A. Chaware, "Transfer learning with fine-tuned MobileNetV2 for diabetic retinopathy," in *Proc. Int. Conf. Emerg. Technol. (INCET)*. India: IEEE, Jun. 2020, pp. 1–4, doi: 10.1109/INCET49848.2020.9154014.

[73] R. Rokhana, W. Herulambang, and R. Indraswari, "Classification of ocular diseases on fundus images using weighted MobileNetV2," in *Proc. Int. Electron. Symp. (IES)*. Indonesia: IEEE, Aug. 2022, pp. 570–575, doi: 10.1109/IES55876.2022.9888652.

[74] S. Tammina, "Transfer learning using VGG-16 with deep convolutional neural network for classifying images," *Int. J. Sci. Res. Publications (IJSRP)*, vol. 9, no. 10, p. p9420, Oct. 2019, doi: 10.29322/ijsrp.9.10.2019.p9420.

[75] M. Mateen, J. Wen, Nasrullah, S. Song, and Z. Huang, "Fundus image classification using VGG-19 architecture with PCA and SVD," *Symmetry*, vol. 11, no. 1, p. 1, Dec. 2018, doi: 10.3390/sym11010001.

[76] X. Zhu, D. Cheng, Z. Zhang, S. Lin, and J. Dai, "An empirical study of spatial attention mechanisms in deep networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6687–6696.

[77] V. Das, S. Dandapat, and P. K. Bora, "Multi-scale deep feature fusion for automated classification of macular pathologies from OCT images," *Biomed. Signal Process. Control*, vol. 54, Sep. 2019, Art. no. 101605, doi: 10.1016/j.bspc.2019.101605.

[78] V. Das, S. Dandapat, and P. K. Bora, "Automated classification of retinal OCT images using a deep multi-scale fusion CNN," *IEEE Sensors J.*, vol. 21, no. 20, pp. 23256–23265, Oct. 2021, doi: 10.1109/JSEN.2021.3108642.

[79] R. Fan, Y. Liu, and R. Zhang, "Multi-scale feature fusion with adaptive weighting for diabetic retinopathy severity classification," *Electronics*, vol. 10, no. 12, p. 1369, Jun. 2021, doi: 10.3390/electronics10121369.

[80] P. Glaret subin and P. Muthukannan, "Optimized convolution neural network based multiple eye disease detection," *Comput. Biol. Med.*, vol. 146, Jul. 2022, Art. no. 105648, doi: 10.1016/j.compbiomed.2022.105648.

MD. ASRAFUL SHARKER NIROB is currently pursuing the degree in computer science and engineering with Daffodil International University, Dhaka, Bangladesh. His research focuses on artificial intelligence, particularly machine learning, deep learning, and explainable AI (XAI), with applications in health informatics and other interdisciplinary domains. As a Dedicated Researcher, he has contributed to the field through multiple published articles and gained hands-on experience as a Research Assistant. His work reflects a strong commitment to advancing AI technologies while ensuring transparency and practical usability in real-world scenarios.

SHOWMICK GUHA PAUL received the bachelor's and master's (M.Sc.) degrees in computer science and engineering from Daffodil International University, Dhaka, Bangladesh. He is currently a Lecturer with the Department of Computer Science and Engineering, Daffodil International University, and the Co-Director of the Health Informatics Research Laboratory (HIRL). He is actively engaged in research in machine learning, deep learning, and data science, with a strong focus on health informatics, AI-driven healthcare solutions, and sustainable computing. He has significant academic and research experience, having been a Research Assistant, and has published several articles in Q1-ranked international journals as a first author. In addition to his research contributions, he serves as a reviewer for various reputed international journals, helping to uphold academic quality and contribute to the advancement of research in his fields of expertise. He is an active member of several research, programming, and technical organizations, promoting interdisciplinary collaboration and innovation. His research interests include machine learning, deep learning, artificial intelligence, computer vision, health informatics, and green computing, with a strong emphasis on their applications in healthcare, sustainability, and intelligent systems development.

ARPA SAHA received the bachelor's and master's (M.Sc.) degrees in computer science and engineering (CSE) from Daffodil International University, Dhaka, Bangladesh. She is currently a Research Associate with the Health Informatics Research Laboratory (HIRL), Daffodil International University. She has built strong expertise in machine learning, deep learning, artificial intelligence, image processing, data mining, natural language processing, health informatics, and computer vision. Her dedication to research is reflected in multiple publications in Q1-ranked international journals. In addition to her research accomplishments, she serves as a reviewer for several reputed international journals, contributing to the academic community by evaluating cutting-edge research. She remains passionate about advancing research in intelligent systems, healthcare technologies, and AI-driven solutions for real-world challenges.

MD ASSADUZZAMAN received the B.Sc. and M.Sc. degrees in computer science and engineering from the Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh. He is currently a Senior Lecturer with the Department of Computer Science and Engineering, Daffodil International University. He is actively engaged in research as a member of the Health Informatics Research Laboratory (HIRL), with a focus on machine learning, deep learning, explainable artificial intelligence (XAI), and federated learning. His work primarily addresses challenges in the healthcare domain, aiming to develop ethical and interpretable AI systems. He has published several research papers in peer-reviewed journals and at international conferences. His academic and research endeavors aim to advance ethical, transparent, and impactful AI solutions for real-world problems.

MD ALAMGIR KABIR (Member, IEEE) received the master's degree in software engineering from Wuhan University, China, and the Ph.D. degree in computer science from the City University of Hong Kong, Hong Kong. He is currently an Inquisitive Researcher. Driven to advance the field of artificial intelligence (AI) and intelligent systems, he brings expertise in software engineering, particularly in software testing, verification, validation, and software metrics. He is in a postdoctoral role with Mälardalen University, Sweden, where he contributed to the Artificial Intelligence and Intelligent Systems Research Group. He is an Assistant Professor with the Department of Computer Science and Engineering, Daffodil International University, Bangladesh. He is committed to developing efficient, explainable, and robust AI techniques. His research has led to numerous publications in top-tier conferences and journals. At Mälardalen University, he focused on creating innovative AI techniques and intelligent systems, which continue to define his research trajectory in fostering advancements within AI and software engineering.

ARIF MAHMUD (Member, IEEE) received the Master's of Science degree with a major in computer engineering from Mid Sweden University, Sweden, and the Ph.D. degree in information systems from the School of Computer Sciences, Universiti Sains Malaysia, Pulau Pinang, Malaysia. Besides, he is employed as an Associate Professor and the Associate Head of the Department of Computer Science and Engineering, Daffodil International University, Bangladesh. He has published numerous academic papers in well-reputed journals and conferences. His current research interests include management information systems, IT operations and management, software engineering, and social networking with communication systems.

ATIQ UR REHMAN received the master's degree in computer engineering from the National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2013, and the Ph.D. degree in computer science and engineering from Hamad Bin Khalifa University, Qatar, in 2019. He was a Postdoctoral Researcher with the College of Science and Engineering, Hamad Bin Khalifa University, from 2019 to 2022. Later, he was an Assistant Professor with the Department of Electrical and Computer Engineering, Pak-Austria Fachhochschule Institute of Applied Sciences and Technology, Pakistan. He was also a Postdoctoral Researcher with the Artificial Intelligence and Intelligent Systems Research Group, School of Innovation, Design, and Engineering, Mälardalen University, Västerås, Sweden. He is currently a Software Engineer with Hamad Bin Khalifa University. His research interests include the development of evolutionary computation, pattern recognition, and machine learning algorithms.

AMINE BERMAK (Fellow, IEEE) received the master's and Ph.D. degrees in electrical and electronic engineering from Paul Sabatier University, France, in 1994 and 1998, respectively. Currently, he is a Professor with the Information and Computing Technology Division, Hamad Bin Khalifa University. He has held various positions in academia and industry in France, U.K., Australia, and Hong Kong. He has published more than 250 articles, designed more than 40 chips, and graduated 14 Ph.D. and 16 M.Phil. students. For his excellence and outstanding contribution to teaching, he was nominated for the 2013 Hong Kong UGC Best Teacher Award (for all HK Universities). He was a recipient of the 2011 University Michael G. Gale Medal for distinguished teaching. He was also a two-time recipient of the "Engineering Teaching Excellence Award" in HKUST, in 2004 and 2009. He has received six distinguished awards, including the Best University Design Contest Award at ASP-DAC 2016, the Best Paper Award at IEEE ISCAS 2010, the 2004 IEEE Chester Sall Award, and the Best Paper Award at the 2005 International Workshop on SOC for Real-Time Applications. He has served on many editorial boards and is an Editor of IEEE Transactions on Biomedical Circuits and Systems and IEEE Transactions on Electron Devices. He is an IEEE Distinguished Lecturer.

Open Access funding provided by 'Qatar National Library' within the CRUI CARE Agreement